

LI-VAD: Lightweight and Interpretable Video Artifact Detection

Laurence Palmer, Ziang Xiao, Haiyi Li, and C.-C. Jay Kuo
University of Southern California, Los Angeles, California, USA
{palmerla, jimmyxia, haiyili, jckuo}@usc.edu

Abstract—In this work, we propose a lightweight and interpretable video artifact detection (LI-VAD) method that leverages natural scene statistics (NSS) and green learning (GL). LI-VAD extracts features from disjoint domains to produce a quality-focused feature vector, which is then supplemented with statistical priors derived from clean training videos. From this set, we aggressively filter out low-quality features and generate additional discriminative features for a prediction module. Subsequent rounds refine the predictions for difficult samples. Experiments demonstrate that our model achieves state-of-the-art artifact detection performance with a fraction of the storage.

Index Terms—Artifact Detection, Green Learning, Machine Learning

I. INTRODUCTION

Video content dominates internet traffic, driven by the growth of social media and video streaming. Consequently, both user-generated content (UGC) and professional-generated content (PGC) must be analyzed for quality to ensure user satisfaction. UGC-BVQA is one of the most active areas of research within video quality assessments due to the proliferation of social media. In this domain, the objective is to regress the mean opinion scores (MOS) aggregated from human ratings. Traditional techniques have relied on natural scene statistics (NSS) and handcrafted features to evaluate video quality. Current research has focused on producing short-form blind video quality assessments (BVQA) using increasingly large deep learning models. For example, recent BVQA models rely on vision transformers and vision language models which require substantial computational resources and lack interpretability.

Video artifact detection (VAD) aims to detect artifacts in a video, typically without reference. It is critical for evaluating and serving video content on large video streaming platforms. Professional streaming services must monitor quality over long time horizons, focusing on detecting artifacts introduced by their content distribution networks that degrade the quality of experience. These domains pose distinct challenges: BVQA requires modeling human perceptual preferences, while artifact detection demands robust identification of degradations across long time horizons. Artifacts are introduced into streaming pipelines during capture, transcoding, or transmission. Detection methods usually predict a single artifact, such as blurring or banding, and do not generalize to out-of-distribution artifacts [3]. Artifacts are correlated with human preferences, but repurposing BVQA-based methods can degrade performance [3] due to domain shift. Furthermore, artifact detection

is particularly important in PGC streaming settings, where efficiency and speed are paramount for multi-hour streams. Finally, multiple artifacts may coexist and confound detectors tailored to a specific set of artifacts.

Motivated by the drawbacks and constraints of existing video artifact detection solutions, we propose a lightweight and interpretable video artifact detection (LI-VAD) method. LI-VAD relies only on handcrafted features and NSS to provide a suite of discriminative features. These features are filtered and supplemented using the green learning (GL) paradigm in a multi-round framework to refine predictions iteratively. We find that our pipeline provides state-of-the-art (SOTA) performance in video artifact detection compared to alternative models. Our contributions are summarized as follows.

- **Feature Extraction Pipeline:** We extract 295 per-frame features organized into spatial, temporal, color, codec, and bitstream branches, each computed at global or local granularity. Experiments demonstrate the importance of a diverse set of features in capturing deviations across distinct domains. Furthermore, we capture feature distributions across a wide range of clean videos to imbue our pipeline with proper inductive biases.
- **GL-based Feature Selection and Generation:** We present a multi-round feature selection and generation tailored to video quality. GL tools such as the Discriminant Feature Test (DFT) [22] and the Least Squares Normal Transform (LNT) [16] are used to identify and generate highly discriminative features in each subsequent round.
- **Dataset for Video Artifact Detection:** We construct a new dataset for VAD, which is generated from LongVideoBench [18]. The dataset contains a variety of long videos and 9 different artifact settings, which are more closely aligned with PGC streaming artifact detection. The dataset is provided at this link.

II. RELATED WORKS

A. Blind Video Quality Assessments

Classical BVQA systems use handcrafted features guided by NSS in conjunction with traditional machine learning methods. VIDEVAL [14] represents the most successful classical approach, leveraging a suite of features aggregated over existing classical approaches. Recently, state-of-the-art BVQA algorithms have relied on larger, more opaque DL architectures. One group is ensembles, including DOVER [20], which

combine scores across various dimensions using deep learning backbones, while models such as MaxVQA [19] leverage large multi-modal models as engines to predict quality and provide explanations for degradation. Despite impressive performance, deep learning approaches incur high computational costs and lack transparency in the pipeline, making them expensive to deploy at scale.

B. Video Artifact Detection

Video artifact detection aims to detect the appearance of a given artifact or a set of degradations without a reference video. Research in video artifact detection has generally focused on single-artifact detection, where a model detects an artifact such as banding [15], rather than a set of artifacts. In production scenarios, diverse artifacts may occur infrequently, making single-artifact detectors unsuitable for broad coverage. On the other hand, VIDMAP [5] is a general artifact detector, which uses minimal convolutional filters to generate impairment maps and predictions. Language-based BVQA algorithms have been adapted to evaluate artifact detection [19], as they score specific artifact dimensions associated with human preferences, such as poor exposure. Recently, MVAD [2] was introduced to detect multiple artifacts simultaneously by extracting artifact-aware features in the spatial domain and using a recurrent neural network architecture to model short- and long-term spatial dependencies. Similar to BVQA models, many of these deep learning pipelines fail to generalize [3], are computationally expensive, and are unexplainable.

C. Green Learning

Green learning (GL) [6] is a learning paradigm that focuses on efficiency and interpretability. Rather than creating globally optimized models as in DL, GL focuses on building modular, lightweight, and interpretable architectures in three separate stages: unsupervised feature extraction, label-guided feature filtering and generation, and final prediction. Unsupervised feature extraction is performed to produce salient representations with methods that include the Saab Transform [7]. These representations are refined using the DFT and RFT [22], which filter out weak features. To supplement the filtered features, the Least Squares Normal Transform (LNT) [16] generates a set of complementary features via linear combinations guided by the label-space. Finally, the filtered and generated features are combined for prediction using classical machine learning. By building in this manner, GL models are well-suited for real-time deployment in resource-constrained environments, thanks to efficiency improvements. Successful applications include medical image segmentation [21] and dehazing [10].

III. PROPOSED LI-VAD METHOD

We pose VAD as a binary classification problem in which positive samples contain artifacts and negative samples are clean. A diagram of the proposed LI-VAD pipeline is shown in Figure 1.

TABLE I
PER-FRAME FEATURE EXTRACTION BRANCHES

Branch	Resolution	Feats	Channels
Spatial Global	Downsampled	44	Y
Spatial Local	Patch	81	Y
Temporal Global	Downsampled, Full	52	Y, Cb, Cr
Temporal Local	Patch, Full	84	Y
Color Global	Downsampled	16	Y, Cb, Cr
Color Local	Full, Patch	6	Y, Cb, Cr
Codec	Full	8	Y, Cb, Cr
Bitstream	—	4	Packet header
Total per frame		295	

A. Input Processing

Given a video V , let Y_i, Cb_i, Cr_i be the corresponding YCbCr channels for a frame at time i . Chroma channels are upsampled to luma resolution using bilinear interpolation. The Y_i, Cb_i , and Cr_i are processed at native resolution and then downsampled. During downsampling, we maintain the original aspect ratio of the video while reducing the minimum axis to 256p. The aspect ratio is preserved during downsampling to avoid geometric distortions that would bias orientation-sensitive features. Certain quality degradations are perceptible at coarse scales, so downsampling may reduce computational load while preserving relevant signals. Global NSS statistics are computed at a downsampled resolution. In contrast, local features and codec features that depend on fine-grained spatial structure are computed over patches or the full-resolution frame. In Table I, we provide a summary of our feature extraction branches and the resolutions they use.

To produce patches, the full-resolution luma channel is partitioned into a $G \times G$ grid. The $P \times P$ patches are extracted from the center points of the cells. In all our implementations, $G = 7$ and $P = 32$, similar to the fragments in FastVQA [17].

B. Feature Extraction

1) *Spatial Features*: Spatial global features are calculated from the downsampled luma channel. We apply the mean-subtracted contrast-normalized (MSCN) transformation [9] given by

$$\hat{Y}_i(k, l) = \frac{Y_i(k, l) - \mu_i(k, l)}{\sigma_i(k, l) + \epsilon}, \quad (1)$$

where $k = 1, \dots, M, l = 1, \dots, N$ are the location of the pixels, $M \times N$ is the frame size, $\mu_i(k, l)$ is the local mean, $\sigma_i(k, l)$ is the local standard deviation after Gaussian blurring, and ϵ avoids division by zero in smooth regions. For the MSCN coefficients, we fit a generalized Gaussian distribution (GGD), extracting shape and variance parameters, skewness, and the variances and kurtoses of the horizontal and vertical paired products, similar to BRISQUE [9]. In addition, we compute mean squared energy from Laws' filter responses [8] over the MSCN coefficients using $25 \times 5 \times 5$ kernels generated from the outer product of 1-D basis vectors ($L5, E5, S5, W5, R5$). These filters were designed for texture analysis, but we found that their responses are useful for VAD. Additional spatial features computed directly on the downsampled luma include

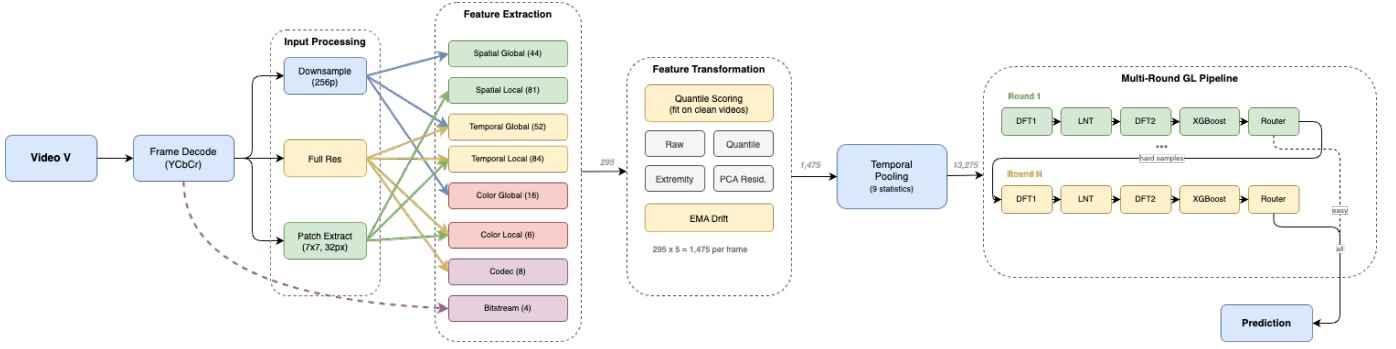


Fig. 1. A high-level overview of the LI-VAD method. Input videos are decoded into YCbCr color planes for feature extraction, transformed via the Quantile Scorer (QS) and EMA drift, temporally pooled, and then processed in the multi-round GL framework.

DCT orientation kurtosis and correlations [12], as well as second-order gradient statistics.

The spatial local branch mirrors the spatial global pipeline, but the features are computed over the native resolution patches described in Section III-A. We compute MSCN variance and kurtosis, Laws’ energy and kurtosis per filter, and Laplacian energy and kurtosis per patch. Given the per-patch feature vectors, we aggregate each metric into three scalars: the mean of the top- K values ($K = 3$) for energy and kurtosis, and the fraction of patches with energy exceeding the median.

2) *Temporal Features*: Spatial, global, and local features capture distortions that are perceptible in the spatial domain. Temporal artifacts, such as stalling, are undetectable without reference frames, which motivates the need for temporal branches.

We capture inter-frame dynamics using normalized frame differences D_i given by

$$D_i(k, l) = \frac{Y_i(k, l) - Y_{i-1}(k, l)}{\sigma(Y_{i-1}) + \epsilon} \quad (2)$$

where Y_{i-1} is the frame immediately preceding Y_i and ϵ avoids division by zero. The temporal global branch applies the same MSCN and NSS pipeline as the spatial branch to D_i , extracting the GGD shape, the variance of the paired product, the Laws’ texture energy, the DCT orientation statistics, and the gradient features of the normalized difference. MSCN coefficient distributions over frame differences in the DCT domain are grounded in NSS as demonstrated by [13], and we extend this finding with Laws’ filters. Additional handcrafted features are included to help parse temporal artifacts from scene cuts and transitions, including absolute differences of frame-level mean luminance.

Temporal local features compute energy and kurtosis in raw patches of D_i , Laws’ energy and kurtosis in MSCN-normalized patches, and Laplacian energy and kurtosis, aggregated via the same metrics as the spatial local branch. Additional features include mean absolute differences (MAD) per patch calculated in D_i . From these per-patch MADs, we compute coefficients of variation and the fraction of patches that exceed the $2\times$ median MAD. Finally, we compute the

maximum per-block MAD across the full-resolution frame using 16×16 blocks.

3) *Color Features*: The spatial and temporal branches operate primarily over the luma, making color-specific artifacts difficult to detect using these branches alone. For example, SDR-to-HDR conversion may cause miscolored skin tones that would be undetectable using only the spatial and temporal branches.

Global color features are computed on downsampled channels across consecutive frames. These capture inter-channel temporal dynamics through chroma-to-luma difference ratios, luma histogram shape, and temporal distance statistics (chi-squared, KL divergence, histogram intersection), as well as chroma distribution measures including per-channel entropy, joint chroma spread, and outlier fraction.

Local color features detect spatial non-uniformity at full resolution through per-block luma and chroma statistics over non-overlapping 32×32 blocks, and per-patch histogram entropy variance over the grid patches described in Section III-A.

4) *Codec and Bitstream Features*: An issue with the previously described branches concerns codec-specific processing. Global downsampling blurs macroblock boundaries, and our patches are not aligned with the block partitions used by codecs such as H.264 and H.265. Consequently, we calculate additional codec features to measure compression artifacts using the full-resolution frame. To capture these artifacts, we calculate horizontal gradient magnitudes at 8- and 16-pixel block boundaries and at non-boundary locations. Then, we calculate ratios across the boundary and non-boundary regions for both block sizes, and the ratio across the two block sizes. We also compute inter-channel correlations to detect decorrelation introduced by lossy codecs. Additional features include the fraction of pixels at extreme values and the ratio of row-to-column mean variances.

The Bitstream features are inspired by ITU P1204.3 [11] and reduced to a simple set. We calculate frame sizes in bits and assign binary labels to I/P/B frames.

C. Feature Transformation and Aggregation

The feature transformation module in our pipeline serves an important purpose: contextualizing raw features relative to

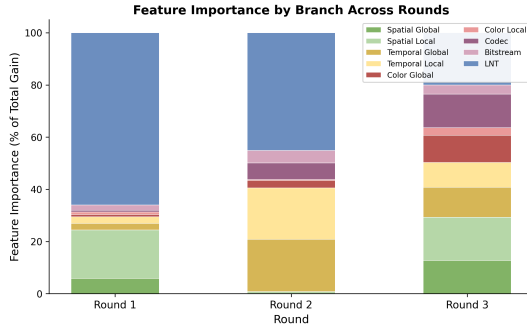


Fig. 2. Feature importance of branches across rounds. LNT and spatial branches dominate the initial round, while codec, temporal, and color features gain importance in later rounds, demonstrating the importance of diverse feature domains.

a reference distribution of clean content and its own recent history. The QS achieves the first objective, while the EMA drift module captures the second.

We calculate the 295 raw features for a sample of clean videos and generate histograms for each. This provides us with a reference distribution for "good" videos. Extreme values or deviations from this baseline align with artifacts. For each raw feature, the QS calculates its quantile q relative to the clean distribution. The extremity measure is defined as

$$E(q) = -\log(2 \min(q, 1 - q)), \quad (3)$$

produce a symmetric two-sided measure relative to the clean median. In addition, we calculate PCA residuals over the full feature vector, using z-scaling parameters and components derived from the clean distribution. The number of components retained for this measure accounts for 95% of the variance in the clean distribution. The input feature vector is projected onto the principal components, and the residual is calculated. The PCA residuals complement quantile and extremity measures by capturing joint anomalies rather than single-feature anomalies. If the dominant covariance structure in the clean videos fails to explain the current frame, then we expect large residual values. In addition, we use EMA to produce a drift vector to track non-stationary features. Finally, we temporally pool the frame-level representations using 9 pooling statistics that capture the distribution shape and temporal dynamics within each window. The final feature vector F is passed to the multi-round pipeline.

D. Multi-round Green Learning

The described feature extraction process corresponds to the unsupervised feature extraction phase of GL. In this section, we discuss the discriminant feature selection, generation, and final prediction pipeline, organized within a multi-round framework. Each subsequent section constitutes a module within one round, as illustrated in Figure 1.

1) *Discriminant Feature Selection:* Given the vector of high-dimensional features F , DFT [22] produces loss scores that capture how discriminative a given feature is in a 1-D space for classification.

For each candidate feature, f_j , and uniformly distributed threshold $c_t \in [\min(f_j), \max(f_j)]$, the training data are partitioned into two subsets:

$$S_{L,j}^t = \{y_i | f_j^i \leq c_t\}, S_{R,j}^t = \{y_i | f_j^i > c_t\}, \quad (4)$$

where y_i are training data samples whose values f_j fall on either side of the partition value. Then, the full loss for f_j with c_t is

$$\ell_j^t = \frac{|S_{L,j}^t| \ell_{L,j}^t + |S_{R,j}^t| \ell_{R,j}^t}{|S_{L,j}^t| + |S_{R,j}^t|}, \quad (5)$$

where $\ell_{L,j}^t, \ell_{R,j}^t$ are the loss values for the left and right subsets. For the DFT, ℓ_j^t is the weighted entropy across the two partitions. The discriminant score for f_j is $\min_t \ell_j^t$ on all t candidate thresholds for the DFT. We rank each feature by this score and retain the top features to produce F_1 where $|F_1| \ll |F|$.

2) *Feature Generation:* To complement the unsupervised discriminative feature set F_1 , we utilize LNT [16]. The LNT takes F_1 and produces an additional set of features using linear combinations guided by labels. The LNT reformulates classification problems as linear regression problems. Suppose that we have L training samples with K classes, where $F_1 \in \mathbb{R}^{N \times L}$ and N is the number of discriminant features selected. Let $\mathcal{T} \in \mathbb{R}^{K \times L}$ where the entry $\tau_{k,l}$ is an indicator variable corresponding to whether the l th training sample belongs to the k th class. The LNT attempts to find an optimal weight matrix $A \in \mathbb{R}^{K \times N}$ that projects F_1 onto the target $\mathcal{T}$ yielding

$$AF_1 + B = \mathcal{T} \quad (6)$$

where B is a rank-one bias to shift means. We remove the bias term, since the shifts do not determine discriminative directions, and solve A using the normal equation $A = \mathcal{T} F_1^T (F_1 F_1^T)^{-1}$. Then, the LNT features are given by $D = AF_1$. In practice, we use an auxiliary XGBoost [1] to group features according to tree paths. The resulting LNT features D are then sent through DFT2, yielding D_1 .

3) *Decision Learning and Routing:* Given D_1 and F_1 , we concatenate them into a single feature vector for prediction using XGBoost [1]. Suppose that we have $P, Y \in \mathbb{R}^L$ where P is the first round prediction vector and Y are the true labels.

Using P , we train an additional XGBoost classifier to detect difficult samples. A difficult sample is defined as one with $|Y - P|$ above a threshold calculated from the residuals, which divides the training samples into two halves. Samples with residuals above the threshold are considered difficult and sent to the next round, while the remaining samples are accepted. The classifier is trained using 5-fold CV, and we use out-of-fold predictions to determine which samples to route to the next training round.

The difficult samples run through an identical pipeline structure as round 1. This allows the subsequent DFT and LNT modules to focus on the hard samples. In addition, subsequent prediction trees are passed cumulative logits as the base margin, so that the gradients are relative to the cumulative

predictions. Thus, subsequent rounds learn corrections on top of previous rounds’ predictions. Figure 2 demonstrates how the set of features evolves through rounds, indicating that our pipeline is identifying discriminant features for hard samples.

IV. EXPERIMENTS

A. Datasets

To construct our internal dataset, we start with the base dataset of LongVideoBench [18]. We chose LongVideoBench as our base video dataset because it contains a large collection of long videos that align with the PGC streaming paradigm. Furthermore, the videos have been screened for quality. LongVideoBench also contains near-PGC and UGC content, making it a diverse dataset for training.

We construct a dataset of 3,670 videos using two duration regimes: 2,204 short videos (15–30s) and 1,466 long videos (> 15m), sampled from LongVideoBench to increase content diversity and evaluate robustness across temporal scales [18]. The resulting dataset contains 1,167 clean videos and 2,505 artifact videos overall; specifically, the short-video subset contains 417 clean and 1,787 artifact videos, while the long-video subset contains 750 clean and 716 artifact videos. Using FFmpeg-based transformations [4], we introduce nine artifact categories with an almost uniform distribution across types: 279 color space mismatch videos, 279 dark-scene videos, 279 graininess videos, 279 interlacing videos, 278 stall/black videos, 279 stall/freeze videos, 279 transcoding videos, 277 transmission-error videos, and 274 mixed-artifact videos. Global artifacts include graininess, generated by injecting temporally varying additive noise; dark-scene degradation, produced by reducing brightness together with mild contrast and saturation suppression; transcoding distortion, obtained by cascading 2–4 generations of H.264/AVC re-encoding over a shuffled bitrate ladder from 600 to 2000 kbps; colorspace mismatch, created by intentionally mis-signaling SDR content as HDR; and interlacing, introduced using a top-field-first interlacing filter. Localized temporal artifacts include transmission errors, synthesized by corrupting an HEVC/H.265 bitstream and compositing the corrupted decode into three random 2–4 s windows, as well as stall/freeze and stall/black events, generated by inserting 1–8 s frozen-frame or black-frame segments. We additionally create mixed-artifact samples by sequentially composing 2–3 randomly selected artifact types from this pool. Note that we use a set of 570 long videos for testing, with 300 clean videos and 30×9 artifact videos.

BVI-Artifact [3] is supplemented with an additional 232 clean clips from the source videos used by BVI-Artifact, due to the lack of artifact-free videos. The clip lengths match those of BVI-Artifact, ranging from 5–10s, and we use ffmpeg [4] to clip the videos with a CRF of 22. The clips are encoded with H. 264 and then transcoded to YUV to match the format of BVI-Artifact videos.

B. Tests and Metrics

Our model is trained solely on our internal training set. The QS is trained on a 1,000-subset of clean videos from the

TABLE II
INTERNAL ARTIFACT DETECTION TEST RESULTS

Artifact Type	LI-VAD	MVAD	MaxVQA
Stall/Black	90.0	93.3	83.3
Stall/Freeze	90.0	93.3	93.3
Transcode	100.0	93.3	93.3
Transmission Error	93.3	93.3	80.0
Grain	100.0	100.0	100.0
Interlace	100.0	96.7	100.0
Dark Scene	100.0	96.7	96.7
Colorspace	100.0	63.3	76.7
Mix	100.0	90.0	100.0
Clean (Acc.)	89.3	33.7	22.7
AUC	0.979	0.698	0.654
F1	0.929	0.688	0.660

TABLE III
ZERO-SHOT ARTIFACT DETECTION AUC ON BVI-ARTIFACT [3]

Artifact Type	LI-VAD	MVAD	MaxVQA
Aliasing	0.930	0.867	0.798
Banding	0.927	0.855	0.718
Black Screen	0.930	0.792	0.682
Blockiness	0.907	0.951	0.873
Dark Scenes	0.948	0.770	0.660
Frame Drop	0.957	0.852	0.727
Graininess	0.960	0.837	0.715
Motion Blur	0.912	0.810	0.775
Spatial Blur	0.931	0.897	0.726
Transmission Error	0.934	0.887	0.677
Per-Type Avg	0.934	0.852	0.735
Overall	0.938	0.804	0.725

wider LongVideoBench dataset, held out from GL pipeline training and disjoint from the validation and testing sets. We use 3 rounds for the artifact detection pipeline, with 300/100, 500/100, and 700/100 splits for the DFT1/DFT2 features, respectively, in each round. The training/validation splits are 0.8/0.2. The window size during training is 5s, with a stride of 5, and localized labels are used to guide window labeling. In addition, we sample at 2 FPS and use an update rate of 0.3 for EMA calculations. In testing, we used the same window size but with a stride of 3. The maximum across all windows is taken as the video-level score.

For comparison, we evaluated MVAD [2] and MaxVQA [19] on the internal test set using the authors’ pre-trained checkpoints. MVAD produces a score on ten artifact dimensions corresponding to those in Table III. We take the maximum to produce a single video score. Similarly, the min score over MaxVQA’s 16-D prediction vector is used to produce a binary label. In addition, we run a zero-shot test on BVI-Artifact using 80 artifact/80 clean samples per artifact. We test MVAD and MaxVQA on these splits with the same binarization scheme as the internal dataset. The results are given in Table II and Table III. For a fair comparison of the methods, we use the threshold that maximizes the binary F1 score for each model on the internal test set. For BVI-Artifact, we report AUC to demonstrate transferability and ranking without enforcing a strict threshold.

C. Results and Analysis

Our results demonstrate the superiority of our method over DL models. For MVAD, recall is high, while clean video accuracy is low. Analyzing their logits, we found that the transmission error and frame drop artifact axes are the most prevalent artifact dimensions in 60.5% and 19.8% of videos, respectively. In particular, the mean and median values of the transmission error logits are 0.709 and 0.805. This aligns with the observations in the training of our pipeline. These two artifacts were the most difficult to parse (frame drops and stalls/freezes) due to slow-moving video and artistic effects. In contrast, MVAD performs well in BVI-Artifact with good AUC across the board. MaxVQA follows a similar trend, with extremely low clean accuracy on our internal dataset and significantly better AUC on BVI-Artifact.

In contrast, our model balances sensitivity and coverage, while generalizing well to out-of-distribution content, artifacts, resolutions, and codecs. BVI-Artifact videos and our clean supplements are sourced in YUV format and 4K resolution compared to H.264/720P in our internal dataset. The relatively low contributions from the codec and bitstream branches in Figure 2, along with the BVI-Artifact results, suggest that our model is not overly dependent on these branches, allowing it to generalize to BVI-artifacts. In particular, our model underperforms MVAD in blockiness.

MaxVQA and MVAD are adapted from their respective domains, and our evaluation procedure differs from multiple artifact detection. Still, we argue that posing detection as a binary classification problem is natural for production scenarios. Furthermore, LI-VAD requires 13.3 MB of storage, while the authors of MVAD report 653 MB for their model and 47.6 MB for MaxVQA [2].

V. CONCLUSION AND FUTURE WORK

LI-VAD provides a novel, lightweight model applicable to video artifact detection. With a significantly smaller model footprint compared to DL baselines, we achieve SOTA performance in VAD. For future work, we hope to adapt our pipeline to include BVQA and prune our feature set and pooling methods for further efficiency gains. We also plan to conduct ablation studies and broader comparisons of models and datasets. Furthermore, some of our features scale linearly with video resolution, which hurts run times on high-resolution videos.

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